Good Looks and the Academic Job Market

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ICEF Applied Economics Conference, October 2026

Beauty Economics

". . . the handsome person can attain all his ends without doing anything."

— Xenophon, Symposium

- Physical attractiveness is a valued and relatively scarce attribute.
- There are wage premia for good looks (and penalties for bad looks) in almost all labor markets.
- Other affected outcomes:
  - employability
  - life satisfaction
  - electoral success
  - cognitive development

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The Measurement Problem

- **Subjectivity**: the same individual is judged differently by different raters.
- **Cultural variation**: raters from different times and places apply different criteria.
- **Prejudice**: ratings are sensitive to ascriptive characteristics of the person being rated (e.g., race, gender, age).
- **Coarse measurement**: most ratings use a 1–5 scale.
- **Irreproducibility**: ratings are difficult to reproduce—both for the original researcher and for others.

Takeaway

Human ratings are noisy, context-dependent, and hard to replicate.

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This Paper

- A new measure of physical attractiveness:
  - **RVE Score** (*Relative Vision Embedding Score*)
- Good properties:
  - Fully reproducible.
  - Computationally efficient and scalable.
  - Near-zero marginal cost.
  - Continuous
- Application: photos of economics job-market candidates.

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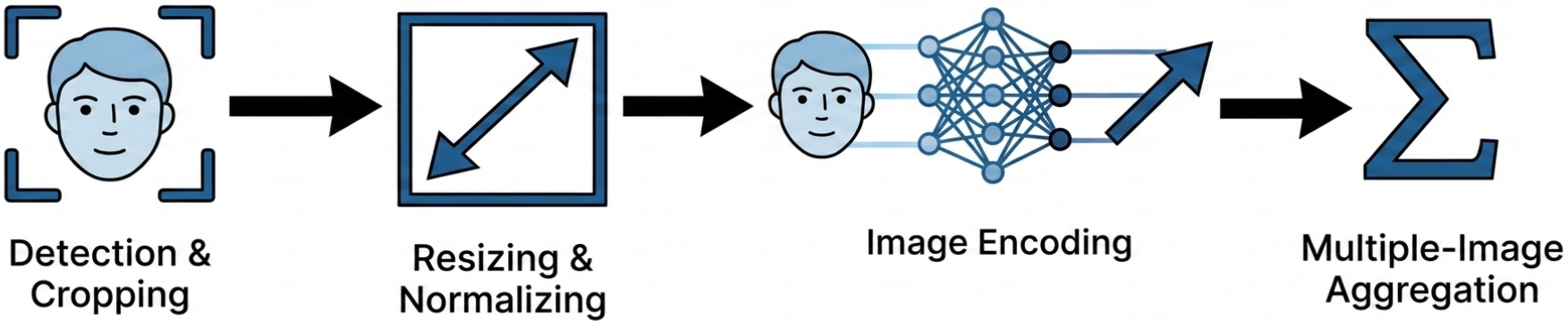
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Related Literature: Measuring Looks

Reference point	Deng et al. (2020) use celebrities as a benchmark for good looks.
Measurement	Peng et al. (2020): random forest on hand-crafted pictorial features. Kosinski (2021): similarity-based analysis using face recognition.
This paper	**RVE Score**: celebrity-relative, no need for training, no feature engineering.

From Image to Vector

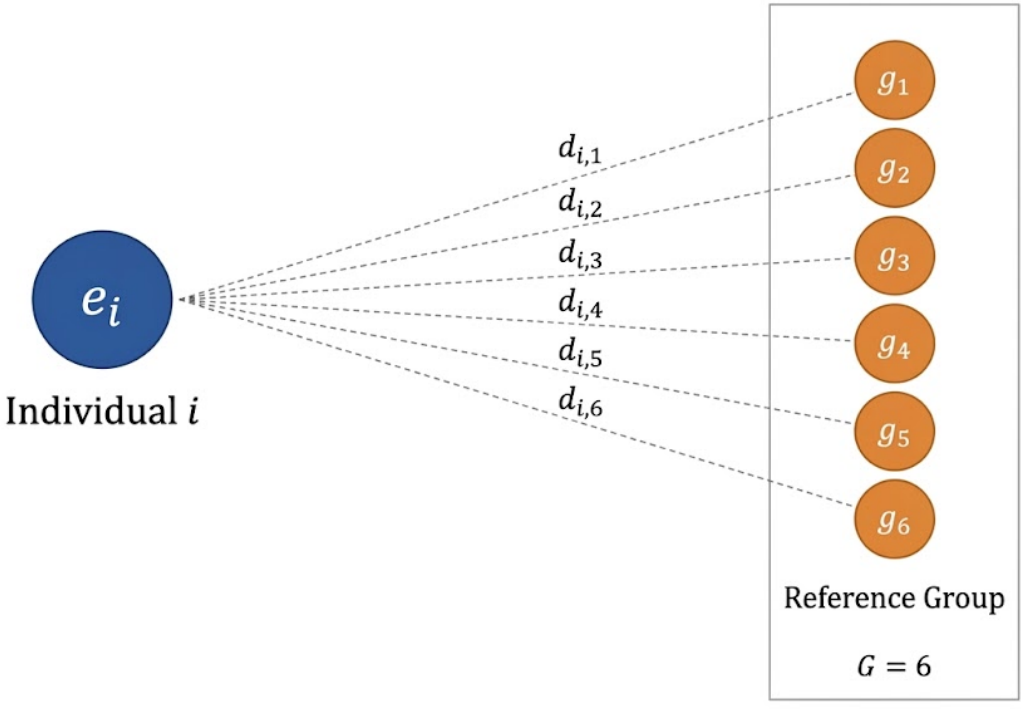


1. `RetinaFace` detects and crops the face on the original photograph.
2. Model-specific preprocessor resizes and normalizes the image.
3. Vision model produces an embedding.
4. Embeddings for multiple photographs are averaged.

Deep Vision Models

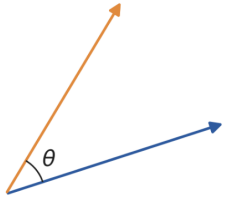
Model	Architecture	Embedding Dimension	Parameters	Source	Developer
DINOv3 ViT-L/16	Vision Transformer	1,024	300M	HuggingFace	Meta AI (2025)
EVA-02 ViT-L	Vision Transformer	1,024	304M	HuggingFace	BAAI (2023)
Qwen3-VL	Multimodal Transformer	2,048	2B	HuggingFace	Alibaba (2025)
Inception v3	Convolutional Neural Network	2,048	24M	Tf Hub	Google (2016)

RVE Beauty Score Calculation



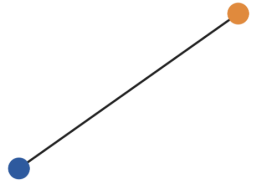
$$b_i^G = \frac{1}{G}\sum_{j=1}^{G} d(e_i, g_j)$$

Cosine similarity



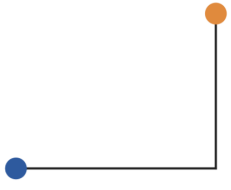
$$\cos(\theta) = \frac{e_i \cdot e_j}{\|e_i\| \, \|e_j\|}$$

Euclidean distance



$$d_E = \sqrt{\sum_k (e_{i,k} - e_{j,k})^2}$$

Manhattan distance



$$d_M = \sum_k |e_{i,k} - e_{j,k}|$$

Reference Group

- Reference group of celebrities
- `www.ranker.com`: rankings based on public user voting
- 776 celebrities in 8 categories (female/male and black/white/hispanic/asian)





Comparison with Human-rated Attractiveness

- The Chicago Face Database 3.0
  - Publicly available
  - 597 individuals with self-reported gender and race identifiers
  - Rated by 28 independent volunteers

CFD Subjects

Top 10


HRS: 5.48


HRS: 5.31


HRS: 5.24


HRS: 5.12


HRS: 5.09


HRS: 5.08


HRS: 5.07


HRS: 4.96

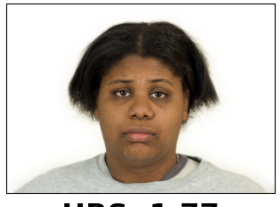

HRS: 4.89


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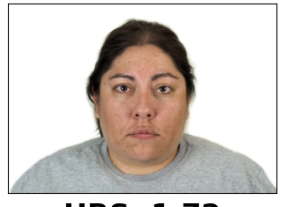
Bottom 10

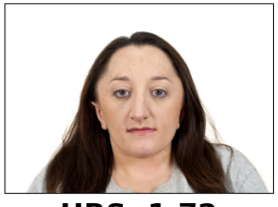

HRS: 1.81

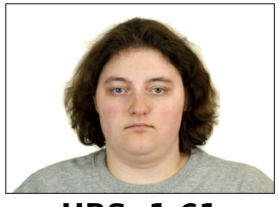

HRS: 1.79

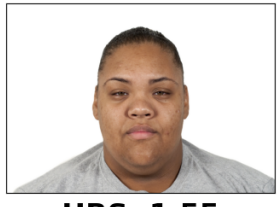

HRS: 1.77


HRS: 1.73

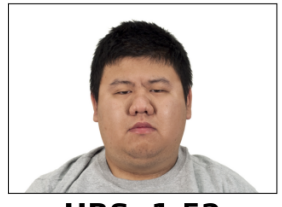

HRS: 1.72


HRS: 1.72


HRS: 1.61


HRS: 1.55


HRS: 1.54

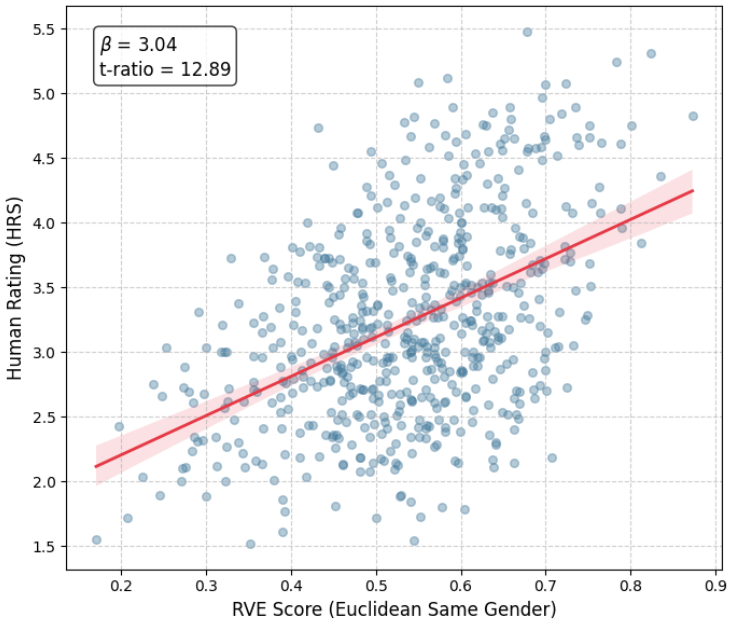

HRS: 1.52

RVE Scores and Human Ratings

		Same Race & Gender			Same Gender			All Celebrities		
Model		Cosine	Euclid.	Manh.	Cosine	Euclid.	Manh.	Cosine	Euclid.	Manh.
DINOv3	ρ	0.140	0.168	0.153	0.135	0.146	0.125	0.359	0.203	0.178
	r	0.147	0.177	0.160	0.137	0.166	0.145	0.385	0.227	0.193
Inception	ρ	0.157	0.289	0.272	0.217	0.334	0.321	0.003	0.129	0.133
	r	0.200	0.295	0.281	0.244	0.330	0.317	−0.015	0.124	0.120
EVA-02	ρ	0.328	0.373	0.385	0.422	0.469	0.484	0.360	0.365	0.389
	r	0.355	0.398	0.410	0.459	0.507	0.519	0.379	0.387	0.405
Qwen3-VL	ρ	0.050	0.081	0.068	0.036	0.064	0.053	0.096	0.090	0.077
	r	0.066	0.096	0.081	0.054	0.083	0.070	0.106	0.098	0.090

Notes: N=597.

Human-rated and RVE Ensemble Scores — CFD



HRS Regression

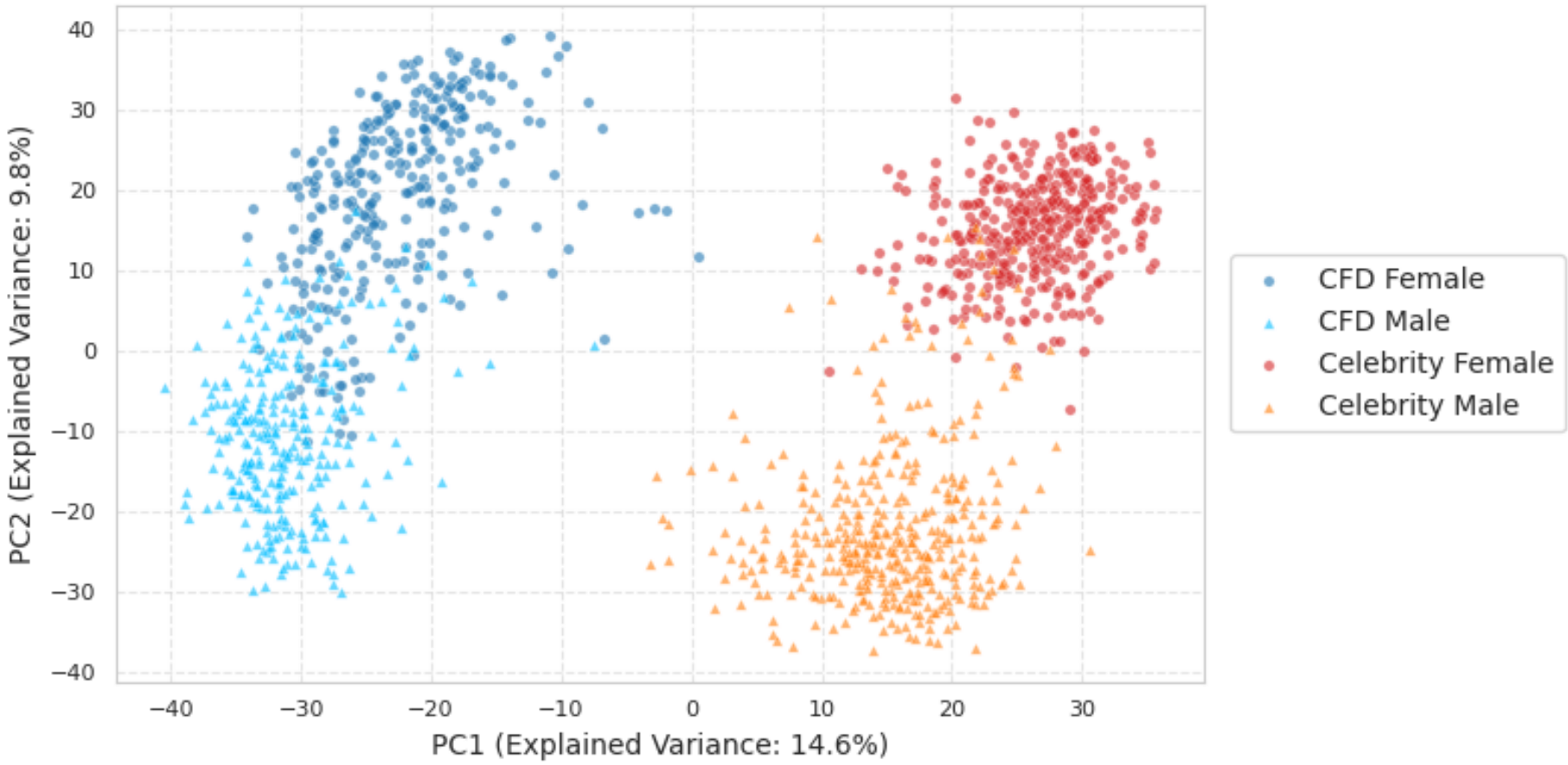
	(1)	(2)	(3)	(4)
RVE Ensemble	3.036***	4.073***	4.284***	4.233***
	(0.216)	(0.188)	(0.190)	(0.192)
Controls				
Gender	✗	✓	✓	✓
Race	✗	✗	✓	✓
Gender × Race	✗	✗	✗	✓

Notes: Dependent variable: CFD human-rated attractiveness (HRS). $N = 597$. Robust standard errors (HC3) in parentheses. * p<.1, ** p<.05, *** p<.01.

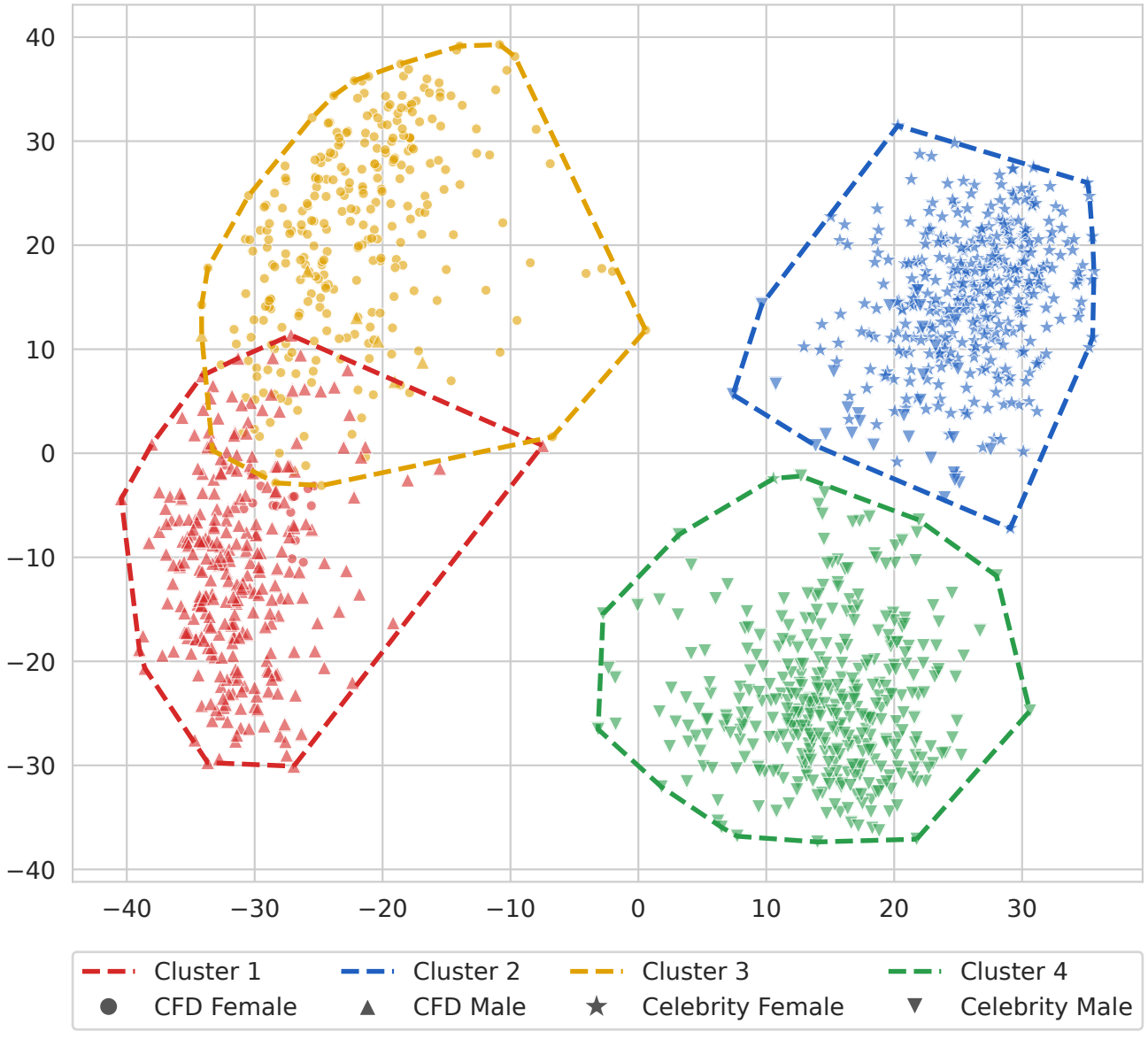
Group-level Mean Scores

Subgroup	*n*	HRS	RVE Ensemble (Euclidean distance)		
			Same gender	Same race/gender	All celebrities
All	597	3.23	0.30	0.19	0.30
Female	307	3.45	0.29	0.15	0.29
Female × Asian	57	3.64	0.30	0.14	0.30
Female × Black	104	3.33	0.26	0.14	0.26
Female × Hispanic	56	3.46	0.29	0.12	0.29
Female × White	90	3.45	0.30	0.17	0.30
Male	290	3.00	0.31	0.24	0.31
Male × Asian	52	2.85	0.27	0.23	0.27
Male × Black	93	3.17	0.31	0.25	0.31
Male × Hispanic	52	2.94	0.31	0.23	0.31
Male × White	93	2.96	0.32	0.26	0.32

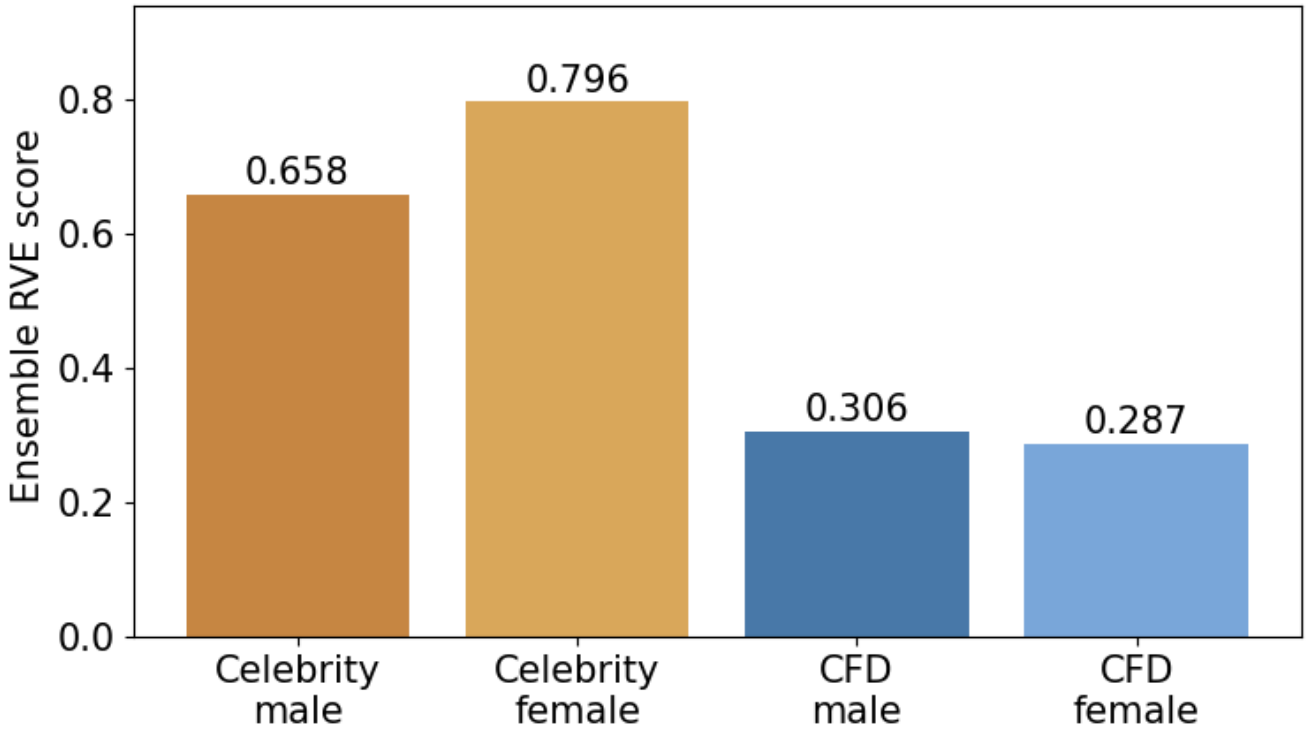
Embedding Visualization (PCA)



K-means Clusterization ($k = 4$)



Average RVE Scores — CFD v. Celebrities



Academic Job Market

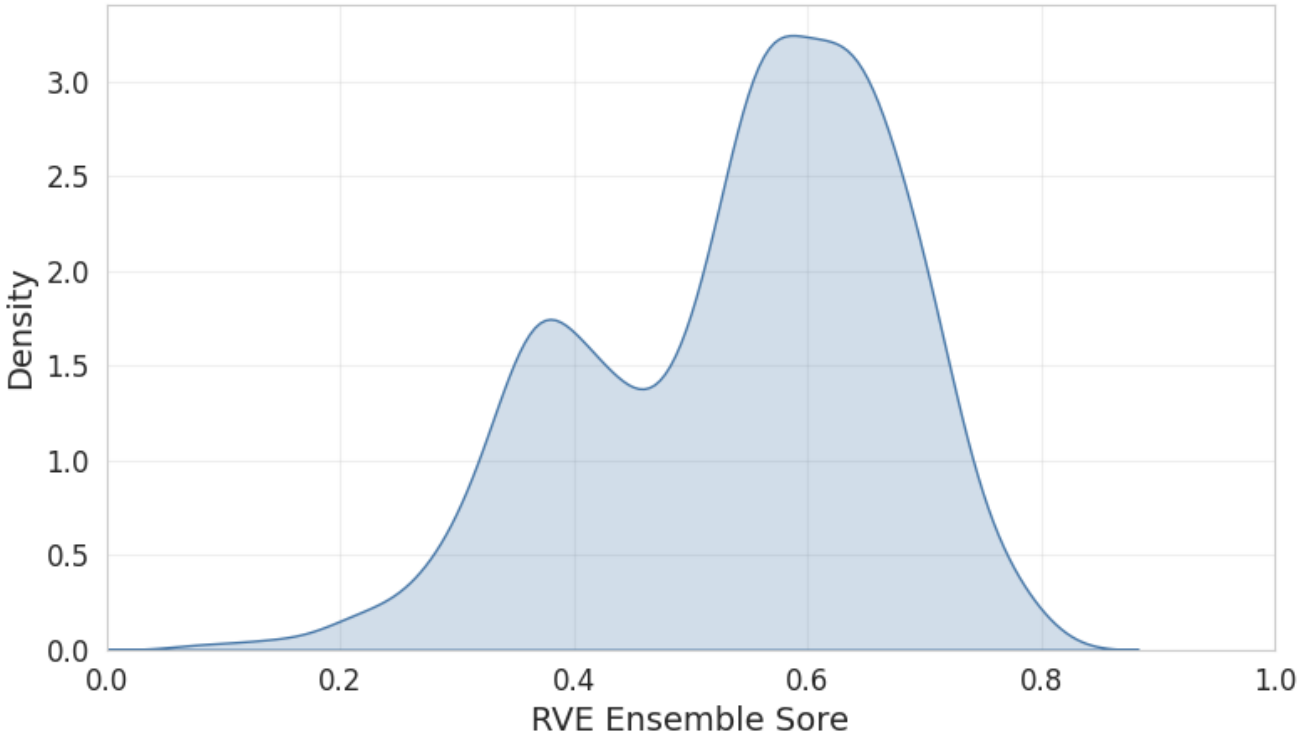
- Applicants to tenure-track position at a Russian research-based economics department
  - 2013–2021
  - Fresh PhDs only
  - Demographic & PhD program characteristics
  - RLs text and letter-writers
  - Job market outcome
  - At least one photo ($N = 630$ /645)

Ensemble RVE Scores

Subgroup	*n*	Mean	Min	Median	Max
All	630	0.54	0.08	0.56	0.80
All Females	165	0.57	0.15	0.57	0.80
Female × Asian	45	0.53	0.19	0.56	0.70
Female × Black	0			NA	
Female × Hispanic	10	0.60	0.49	0.59	0.71
Female × White	110	0.58	0.15	0.59	0.80
All Males	465	0.53	0.08	0.56	0.79
Male × Asian	133	0.47	0.08	0.48	0.76
Male × Black	5	0.51	0.39	0.52	0.62
Male × Hispanic	58	0.57	0.26	0.58	0.79
Male × White	269	0.56	0.13	0.58	0.79

Notes: Job market candidates sample. The RVE scores are calculated using Euclidean distance and same-gender celebrities as a reference group.

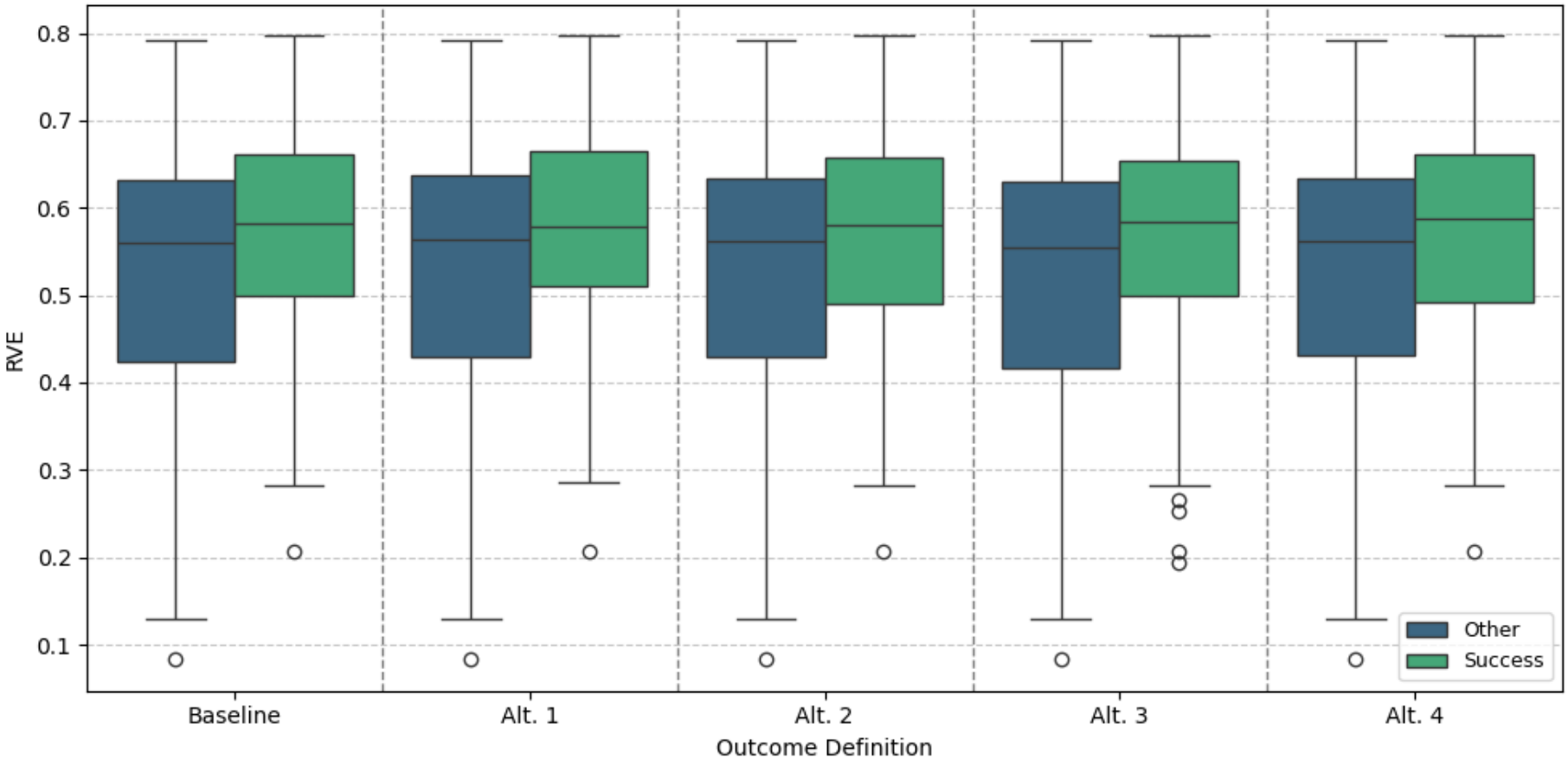
RVE Scores — Kernel Density Estimate



Job Market Outcome Definitions

Definition	Criteria	Success Rate
BASELINE	Top-100 Academic OR Top-50 Policy	27.6%
ALT. 1	Same as Baseline MINUS Post-Docs & Teaching	18.1%
ALT. 2	Baseline PLUS Top Tech Companies	29.8%
ALT. 3	Top-150 Academic OR Top-75 Policy	38.3%
ALT. 4	Top-75 Academic OR Top-35 Policy	23.7%

RVE and Job Market Outcomes



Regression Results — Baseline

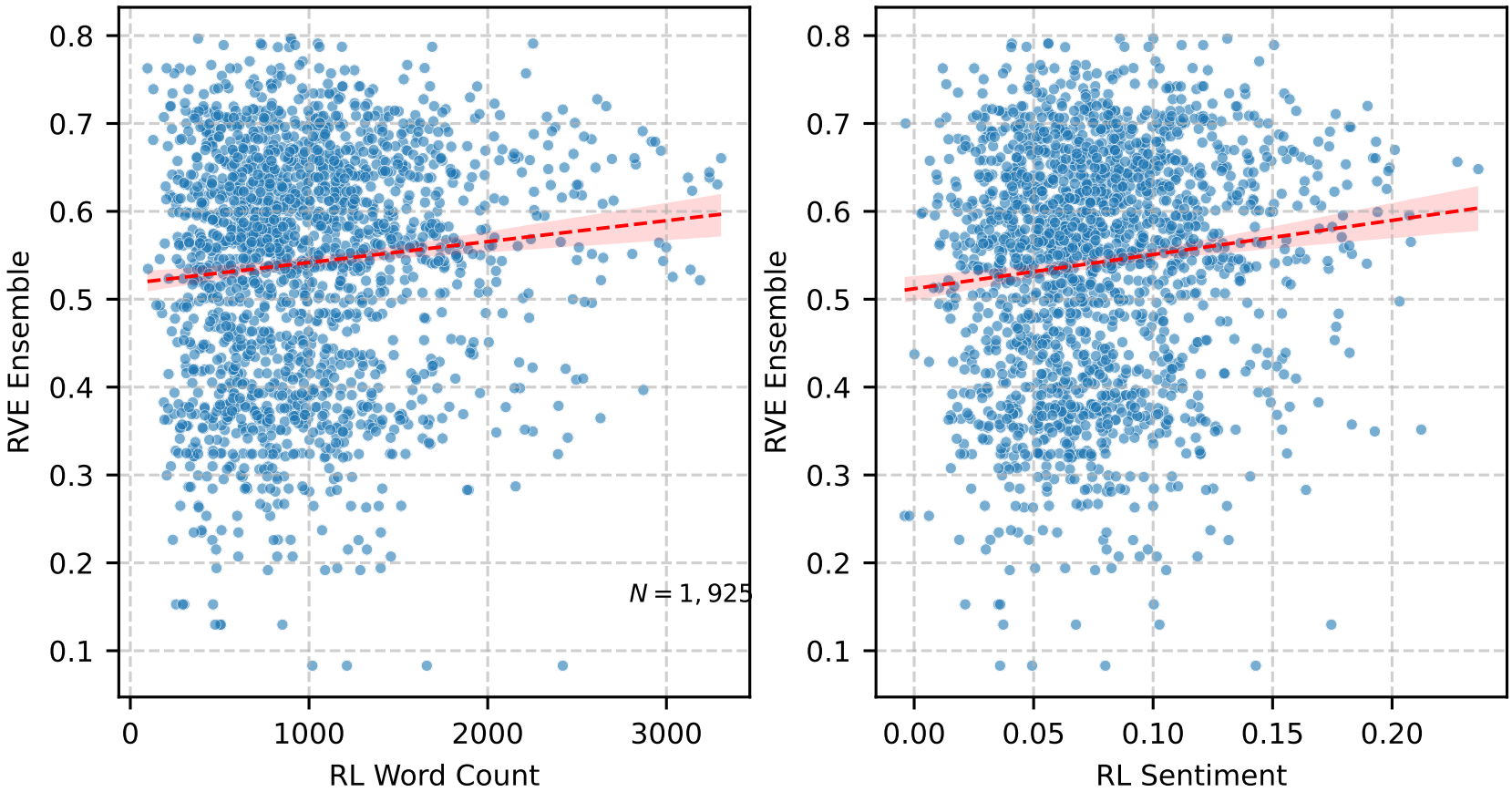
	Outcome: Top-100 Academic or Top-50 Policy					
	(1)	(2)	(3)	(4)	(5)	(6)
RVE Ensemble	0.431***	0.308**	0.251*	0.185	0.185	0.152
	(0.127)	(0.138)	(0.144)	(0.142)	(0.141)	(0.141)
RL Avg. Length (,000 words)				0.280***		0.233***
				(0.052)		(0.054)
RL Avg. Sentiment (pp)					0.031***	0.021***
					(0.007)	(0.007)
Adj. R^2	0.014	0.043	0.072	0.119	0.101	0.130
Controls:						
Candidate	✗	✓	✓	✓	✓	✓
PhD Program	✗	✗	✓	✓	✓	✓
Year	✗	✗	✓	✓	✓	✓
RL Length	✗	✗	✗	✓	✗	✓
RL Sentiment	✗	✗	✗	✗	✓	✓

Notes: $N = 630$. HC3 standard errors in parenthesis.

Regression Results — All Outcomes

	Outcome				
	Baseline	Alt. 1	Alt. 2	Alt. 3	Alt. 4
RVE Ensemble	0.152	0.137	0.122	0.226	0.091
	(0.141)	(0.125)	(0.147)	(0.156)	(0.136)
RL Avg. Length (,000 words)	0.233***	0.180***	0.216***	0.251***	0.178***
	(0.054)	(0.052)	(0.055)	(0.057)	(0.053)
RL Avg. Sentiment (pp)	0.021***	0.015**	0.023***	0.024***	0.022***
	(0.007)	(0.006)	(0.007)	(0.008)	(0.007)
Adj. R^2	0.130	0.129	0.126	0.143	0.110
Controls:					
Candidate	✓	✓	✓	✓	✓
PhD Program	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓
RL Length	✓	✓	✓	✓	✓
RL Sentiment	✓	✓	✓	✓	✓

Notes: $N = 630$. HC3 standard errors in parenthesis.


RVE and RL Characteristics
RVE Ensemble
RL Word Count
N = 1,925
RVE Ensemble
RL Sentiment

Letter Writers' Characteristics

	Mean	Min	Max
Adviser	0.310	0.0	1.0
Male	0.881	0.0	1.0
Experience	19.41	1.0	57.0
Top 10 RePEc	0.514	0.0	1.0
Top 5 RePEc	0.348	0.0	1.0
Full Professor	0.610	0.0	1.0

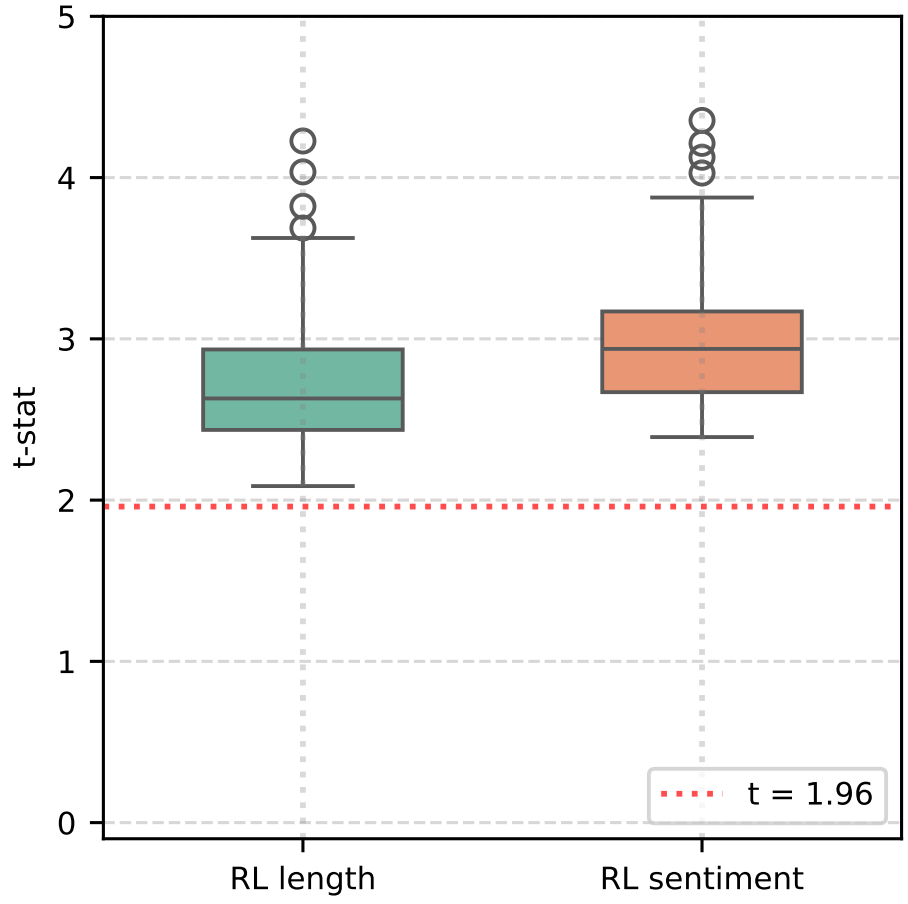
$N = 1925$

Regression Results — RLs Outcomes

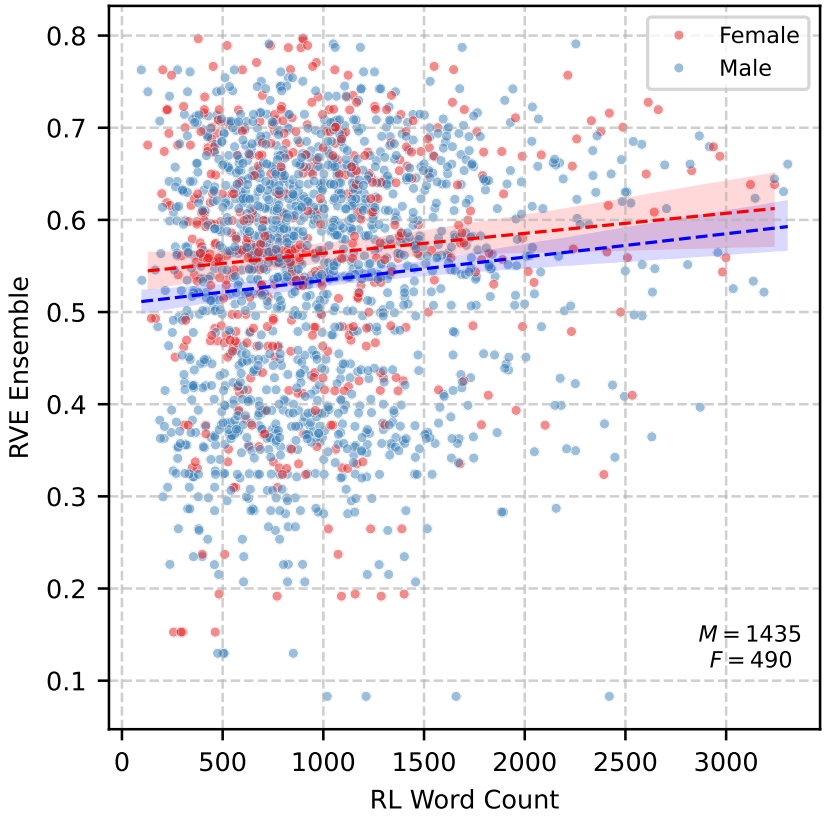
	(1)	(2)	(3)	(4)	(5)
Outcome: RL Word Count					
RVE Ensemble	393.0***	265.2**	252.2**	254.3**	241.6**
	(108.6)	(107.2)	(103.5)	(100.4)	(100.1)
Outcome: RL Sentiment					
RVE Ensemble	0.030***	0.022***	0.020***	0.020***	0.019**
	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
Controls:					
Candidate	✗	✓	✓	✓	✓
Year	✗	✓	✓	✓	✓
PhD Program	✗	✗	✓	✓	✓
Writer Char.	✗	✗	✗	✓	✓
Writer Rank	✗	✗	✗	✗	✓

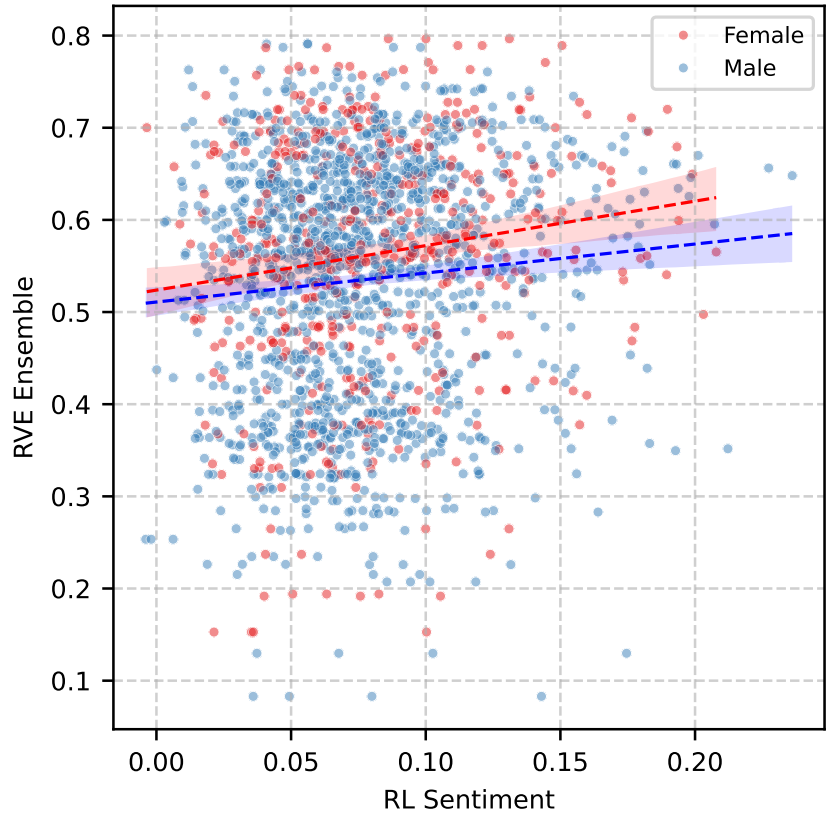
Notes: Euclidean metric and same-gender ref. group. $N = 1,925$. Candidate-level clustered standard errors in parenthesis.

Alternative Measures/Specifications (45)



RVE and RL Characteristics



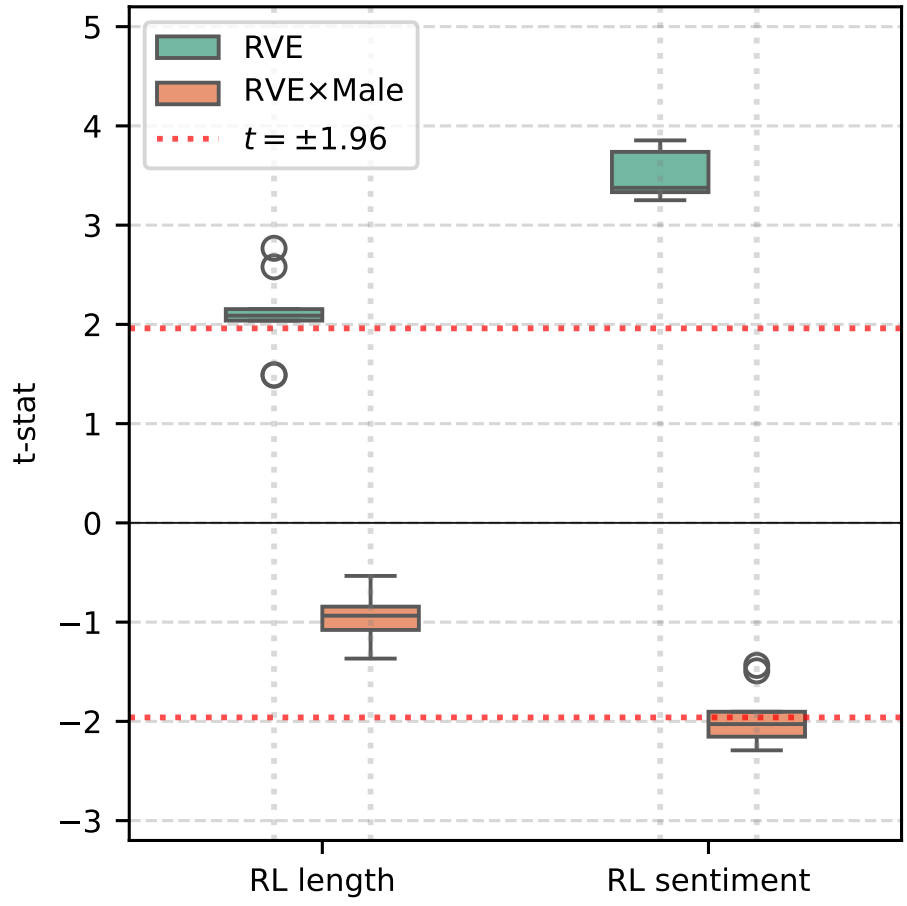


Regression Results — RLs Outcomes by Gender

	Outcome	
	RL Length	RL Sentiment
Male	126.9	0.011
	(123.5)	(0.009)
RVE Ensemble	399.9**	0.044***
	(191.5)	(0.013)
RVE Ensemble×Male	-212.5	-0.033**
	(219.3)	(0.016)
Controls:		
Candidate	✓	✓
Year	✓	✓
PhD Program	✓	✓
Writer Char.	✓	✓
Writer Rank	✓	✓

Notes: Euclidean metric and same-gender ref. group. $N = 1,925$. Candidate-level clustered standard errors in parenthesis.

Alternative Measures/Specifications (9)



Conclusion: A New Measure of Looks

- The **RVE Score** (*Relative Vision Embedding Score*):
  - Distance from a candidate's face to a reference group of celebrities, in a deep-vision embedding space.
- Properties:
  - Fully reproducible.
  - Continuous, not a coarse 1–5 scale.
  - No training or hand-crafted features.
  - Near-zero marginal cost.
- Validated against human attractiveness ratings (Chicago Face Database).

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- Properties:
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- **Finding 1.** Higher RVE Score ⇒ better job-market outcomes.
  - Positive but not completely robust.
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Good Looks and the Academic Job Market

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ICEF Applied Economics Conference, October 2026

References I

Deng, W.; Li, D., and Zhou, D. Beauty and Job Accessibility: New Evidence from a Field Experiment. *Journal of Population Economics*, 33(4):1303–1341, 2020.

Kosinski, M. Facial Recognition Technology Can Expose Political Orientation from Naturalistic Facial Images. *Scientific Reports*, 11:100, 2021.

Peng, L.; Cui, G.; Chung, Y., and Zheng, W. The Faces of Success: Beauty and Ugliness Premiums in E-Commerce Platforms. *Journal of Marketing*, 84(4):67–85, 2020.